Analisa A. Milkey
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amilkey1.github.io
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Current
Postdoctoral Scholar, Division of HIV, Infectious Diseases, and Global Medicine
University of California, San Francisco
Advisor: Dr. Nicola Mueller

Education
Doctor of Philosophy, Ecology & Evolutionary Biology August 2026
University of Connecticut
GPA 4.0
Advisor: Dr. Paul Lewis
Dissertation title: Sequential Monte Carlo Applications in Phylogenetics

Bachelor of Science, Biological Sciences June 2020
University of California, Davis
University Honors Program
GPA 3.88
Honors thesis advisor: Dr. Peter Wainwright

Skills
Experience with C++, R, Python, bash, git
Experience with RevBayes and BEAST software.
Experience with Access and Symbiota databases.

Publications [+]denotes undergraduate mentee
Milkey, A., Chen, J.[+], and Lewis, P. O. (2026). How much information is there for inferring species trees? *bioRxiv*, 2026.04.01.715836.

Milkey, A. and Lewis, P.O. (2026). Estimating Bayesian phylogenetic information content using geodesic distances. *bioRxiv,* 2026.03.31.715656.

Li, A., Liu, P. Y., Wang, Y. B., **Milkey, A.**, Lewis, P., & Chen, M. H. (2026). A comparison of Monte Carlo based marginal likelihood estimators. *Wiley Interdisciplinary Reviews: Computational Statistics, 18*(1), e70058.

Milkey, A., Chen, M. H., Wang, Y. B., Li, A., & Lewis, P. (2025). The sequential multispecies coalescent. *bioRxiv*, 2025.01.31.635964. (In revision at *Systematic Biology.*)

Wang, Y. B., **Milkey, A**., Li, A., Chen, M. H., Kuo, L., & Lewis, P. O. (2023). LoRaD: Marginal likelihood estimation with haste (but no waste). *Systematic Biology*, *72*(3), 639-648. https://doi.org/10.1093/sysbio/syad007

Hodge, J. R., Song, Y., Wightman, M. A., **Milkey, A.**, Tran, B., Stajner, A., Roberts, A. S., Hemingson, C. R., Wainwright, P. C., & Price, S. A. (2021). Constraints on the ecomorphological convergence of zooplanktivorous butterflyfishes. *Integrative Organismal Biology, 3*(1), obab014. https://doi.org/10.1093/iob/obab014

Software [+]denotes undergraduate mentee
Milkey A., Korte E.[+], Lewis P.O. (2023). *lorad: Lowest Radial Distance Method of Marginal Likelihood Estimation*. R package version 0.0.1.0, https://CRAN.R-project.org/package=lorad.

Invited Presentations [+]denotes undergraduate mentee
Milkey, A., Lewis., P. Using geodesic tree distances to estimate Bayesian phylogenetic information content. Oral presentation at Society for Industrial and Applied Mathematics, New York-New Jersey-Pennsylvania section, 2026 meeting. New Brunswick, NJ. October 2026.

Milkey, A., Chen, J.[+], Lewis, P. How much information is there about the species tree even if gene trees are known without error? Invited talk at Society of Systematic Biologists symposium: Machine learning and new inference algorithms: expanding what is possible in evolutionary biology and phylogenetic analysis. Oral presentation at Evolution 2025 meeting (Annual joint meeting of the American Society of Naturalists, the Society for the Study of Evolution, and the Society of Systematic Biologists.) Athens, GA. June 2025.

Contributed Presentations *denotes equal contribution
Milkey, A., Lewis., P. Using geodesic tree distances to estimate Bayesian phylogenetic information content. Oral presentation at UConn Ecology and Evolutionary Biology Graduate Student Symposium, Storrs, CT. February 2026.

Milkey, A., Lewis, P. How much information is there about the species tree even if gene trees are known without error? Oral speed talk at UConn Ecology and Evolutionary Biology Graduate Student Symposium, Storrs, CT. March 2025.

Milkey, A., Lewis, P. Testing the sequential multispecies coalescent. Oral presentation at Evolution 2024 meeting (Annual joint meeting of the American Society of Naturalists, the Society for the Study of Evolution, and the Society of Systematic Biologists.) Montreal, Canada. July 2024.

Milkey, A., Lewis, P. Testing the sequential multispecies coalescent. Oral speed talk at UConn Ecology and Evolutionary Biology Graduate Student Symposium, Storrs, CT. February 2024.

Milkey, A., Lewis, P. The sequential multispecies coalescent. Oral presentation at Evolution 2023 meeting (Annual joint meeting of the American Society of Naturalists, the Society for the Study of Evolution, and the Society of Systematic Biologists.) Albuquerque, NM. June 2023.

Milkey, A., Lewis, P. The sequential multispecies coalescent. Oral presentation at UConn Ecology and Evolutionary Biology Graduate Student Symposium, Storrs, CT. February 2023.

Wang, Y. B., **Milkey, A.,** Li, A., Chen, M., Kuo, L., Lewis, P. LoRaD: marginal likelihood from a single posterior sample. Poster presentation at Evolution 2022 meeting (Annual joint meeting of the American Society of Naturalists, the Society for the Study of Evolution, and the Society of Systematic Biologists.) Cleveland, OH. June 2022.

Wang, Y. B., **Milkey, A.,** Li, A., Chen, M., Kuo, L., Lewis, P. LoRaD method for marginal likelihood estimation. Oral presentation delivered on zoom at UConn Ecology and Evolutionary Biology Graduate Student Symposium, Storrs, CT. February 2022.

Milkey, A., Hodge, J., Wainwright, P.C. Coral feeding shapes behavior in butterflyfishes (Chaetodontidae). Oral presentation delivered through video at UC Davis Undergraduate Research Conference, Davis, CA. April 2020.

Deshmukh*, R., **Milkey*, A.** The effects of early home environment on young Mexican American mothers' ethnic identity. Oral presentation delivered through video at Western Regional Psychology Association Conference, San Francisco, CA. October 2020.

Leung*, T., **Milkey*, A.,** Rusit*, X. Elongation in Eupercaria. Oral presentation delivered at UC Davis Undergraduate Research Conference, Davis, CA. April 2018.

Awards

EEB Merit Award ($750)	April 2026
Halloran Scholarship, Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID) (~$1500)	July 2026
Summer Doctoral Dissertation Fellowship ($2000)	April 2026
The EEB Graduate Student Fund to the Department of Ecology and Evolutionary Biology and Connecticut State Museum of Natural History ($494)	April 2025
Schaefer Fund and the CBC Endowment to the Department of Ecology and Evolutionary Biology and Connecticut State Museum of Natural History ($1215)	April 2024
EEB Headship Award ($500)	May 2023
The Bamford Fund to the Department of Ecology and Evolutionary Biology and Connecticut State Museum of Natural History ($520)	May 2023
NSF Graduate Research Fellowship ($147,000)	Aug 2023 - July 2026
The Ecology and Evolutionary Biology Graduate Student Research Endowment Fund for Research to the Department of Ecology and Evolutionary Biology ($1245)	April 2022
Department Citation, UC Davis College of Biological Sciences	June 2020
Law Family Award, UC Davis Herbarium ($250)	April 2020
Goldwater Scholarship ($7500)	April 2019

UC Davis Regents' Scholarship ($30,000) Sept 2016 - June 2020

Teaching Experience

Teaching Assistant; Workshop on Molecular Evolution May 2023, 2024
Graduate Teaching Assistant; University of Connecticut
- Lab Instructor, EEB5249, Phylogenetics Jan 2024 - May 2024
- Teaching Assistant, EEB2245, Evolutionary Biology Jan 2023 - May 2023
- Lab Instructor, BIOL1108, Principles of Biology II Aug 2021 - Dec 2022

College Essay Writing Tutor; The Princeton Review June 2020 - July 2021
Teaching Assistant; Greentech Education and Employment March 2018 - Feb 2019

Service

UConn EEB NSF GRFP Peer Reviewer 2025
Letters to a Pre-Scientist, STEM professional 2023 - present
UConn EEB Graduate Student Outreach Committee 2022 - 2026
Natchaug Elementary School Guest Scientist 2022 - 2026

Other Experience

Curatorial Assistant; UC Davis Herbarium April 2018 - June 2021
Driver; Yolo Food Bank March 2020 - June 2020

Academic Organizations

Phi Beta Kappa Honor Society April 2019 - present
UC Davis Regents' Scholars Society Sept 2016 - June 2020